Bakelit – manual

Falk Instruments · version 1.8

Bakelit is a polyphonic analog-style synthesizer with three oscillators, subharmonics, two filter types, a step sequencer, an arpeggiator and polyrhythms. This manual takes you from installation to your own sound in ten minutes.

The panel labels are in Swedish with English in parentheses, e.g. *FILTERKURVA (FILTER ENV)*. Knob names are in English.

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1. Install

Mac (VST3 + AU)

1. Unzip `Bakelit-mac.zip`.
2. Right-click **Installera.command** → **Open** (first time only, as the app is not yet notarized).
3. Restart your DAW. Bakelit appears under *Falk Instruments*.

Windows (VST3)

1. Unzip `Bakelit-windows.zip`.
2. Double-click **Installera.bat** and accept the administrator prompt.
3. Restart your DAW.

Reason: find Bakelit in the browser under *Instruments → Plugins → Falk Instruments*. Also works in Ableton, Logic (AU), Cubase, FL Studio, Bitwig, Studio One and Reaper.

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2. Up and running in a minute

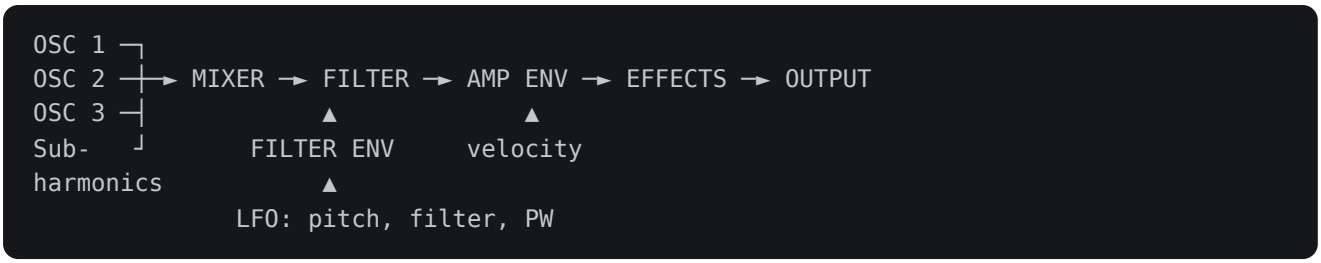
1. Put Bakelit on an instrument track and play.
2. Browse sounds with **<** and **>** at the top right, or click the name to see all sounds by category.
3. Try the **Sequence**, **Polyrhythm** and **Subharmonic** categories – hold a key and press Play in your DAW.
4. Made something you like? **Spara (Save)**.

Good to know

- **Double-click** a knob = default value.
- Drag up/down on a knob to turn it. The value is shown while dragging.
- Hover over a control for a short explanation.
- Drag the bottom-right corner of the window to resize it.
- **Sekv. (Seq)** and **Klav. (Keys)** fold out the sequencer and keyboard. The lamp on Sekv. lights when the sequencer is playing while folded in.

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3. Signal flow



4. The sections

Oscillators (OSC 1–3)

OSC 1 is the reference. It plays exactly the note you press and sets the register with **Octave** (−2 to +2). **OSC 2 and 3 are tuned against OSC 1**: **Semi** sets intervals (+7 = fifth, ±12 = octave), **Fine** detunes slightly for a fat, beating sound.

Knob	Does
Wave	Saw, Square, Triangle, Sine
PW	Pulse width for Square (50 % = square, narrower = thinner, nasal)
Sub N Div	Subharmonic divider: the oscillator's frequency ÷ 1–16
Sub N	Subharmonic level (0 = off)

Subharmonics (UNDERTON / Sub 1–3) are framed inside each oscillator. They follow their oscillator's pitch, waveform and PW exactly. ÷2 = one octave down, ÷3 = a fifth below that, ÷4 = two octaves down, ÷5, ÷6 … give ever darker, stranger tones. Try OSC 1 with Sub 1 Div = 3 and Sub 1 = 50 %.

Mixer (NIVÅER)

Levels for **OSC 1–3**, **Sub Oct** (a classic square one octave below OSC 1) and **Noise**.

Filter (VCF)

Knob	Does
Type	**Ladder** (round, fat, 24 dB) · **Steppe LP** (aggressive, raw) · **Steppe BP** (band-pass, nasal)
Cutoff	Cutoff frequency – dark/bright
Reso	Resonance – squeal at the cutoff; self-oscillates at the top
Drive	Overdrive into the filter
Env	How much the filter envelope opens the filter
Key	Filter follows the keyboard (brighter higher up)
Velocity	Harder playing = brighter
Feedback	Filter output fed back in – dirtier, thicker
Warmth	Soft saturation after the filter

Filter env and Amp env (FILTERKURVA / VOLYMKURVA, ADSR)

Attack, **Decay**, **Sustain** (level while the key is held), **Release** (tail after letting go). The amp envelope's **Velocity** sets how much playing strength affects volume.

Modulation (LFO)

A slow oscillator that moves the sound: **Wave**, **Rate**, **Fade In** (vibrato fades in), and amounts to **Pitch** (vibrato), **Filter** (wah) and **PW** (pulse-width modulation, all three oscillators). The mod wheel adds extra vibrato.

Voice (STÄMMA)

Knob	Does
Mode	**Poly** (chords) · **Mono** · **Legato** (glide only on tied notes)
Glide	Portamento between notes
Bend	Pitch bend range
Drift	Analog instability in tuning and filter
Spread	Spreads voices across the stereo field
Unison	1–4 layers per voice
Detune	How far the unison layers are detuned

Effects and Output (EFFEKTER / UTGÅNG)

Chorus (Off/I/II), **Delay** (Time, Feedback, amount), **Reverb** (amount). **Volume** is the master level. *Stämmor* (voices) at the top right shows how many voices are playing.

5. Step sequencer and arpeggiator

Fold out with **Sekv. (Seq)**. Follows your DAW's tempo; without transport it runs on its own clock while you hold a key.

Mode

- **Arp** – plays the chord you hold. **Arp** sets the order (Up, Down, Up/Down, Random, As Played), **Octaves** 1–4.
- **Seq** – plays the pattern, transposed by the key you hold.

The grid (click/drag): **Step** on/off · **Accent** · **Tie** (tie to next step = glide) · **Pitch** (drag up/down, ±24) · **Prob** (probability) · **Gate** (note length). Double-click resets Pitch/Prob/Gate.

Control	Does
Rate	Step length (1/4 … 1/32, triplets, dotted)
Swing	Delays every other step
Gate	Length of all notes
Steps / Acc steps / Gate steps	Separate lengths (1–32) → polymeter, e.g. 7 steps against 4/4
Latch	Keeps playing after you release the keys
MIDI out	Sends the notes to your DAW (not Reason/Logic – drag MIDI instead)
Euclid	Spreads *Hits* evenly over the steps. **Init** resets the pattern

Drag MIDI to DAW (DRA MIDI TILL DAW) – drag the box to a MIDI track to get exactly what is playing as a MIDI clip. The length comes from your DAW's loop locators or from **Från/Till** (from/to bar). The sequencer then switches off to avoid doubled notes; **Slå på (Re-enable)** turns it back on.

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6. Polyrhythm (POLYRYTM)

Turn on **Polyrhythm** in the sequencer row. It takes over from Arp/Seq and plays from the key you hold. The rings show where the hits fall – where lanes meet you get an **accent**.

Four lanes, each with:

Column	Does
Lane (1–4)	Lane on/off
Type	**Bar** = *Pulses* hits spread evenly over the cycle · **Div** = one hit every *Pulses* steps of Rate (clock divider)
Pulses	Hits per cycle (or the divider in Div)
Hits	How many of the pulses play, evenly spread (Euclidean)
Pitch	Lane pitch relative to the key, ±24 semitones
Steps + Notes	1–8 pitches the lane steps through, one per hit. Drag the cells

Cycle bars (1–4) = cycle length in bars for Bar lanes.

Quick setups: 4:3, **3:2**, **5:4**, **7:4**, **Div** (÷2 ÷3 ÷5 ÷7), **Init**.

Get started: click **4:3**, hold a note and press Play. Then change *Pulses* on lane 2 to 5, or switch on lane 3.

7. Your own sounds

- **Spara (Save)** stores all knobs and patterns. Same name = overwrite (asks first).
- **Ta bort (Delete)** removes the selected user sound.
- User sounds are at the top of the list under **Egna (User)**.
- The files (`.bakelit`) live in **Documents/Falk Instruments/Bakelit/Presets** – copy the folder to back up or to move between Mac and Windows.
- Your DAW project remembers everything, including patterns and folded-out panels.

8. Troubleshooting

Problem	Fix
Not showing in the DAW	Restart the DAW. Run the installer again. Let the DAW rescan plugins
Mac: “cannot be opened”	Right-click → Open, or System Settings → Privacy & Security → Open Anyway
Sequencer silent	Mode or Polyrhythm must be on, and a key held (or Latch)
Doubled notes after dragging MIDI	The sequencer should be off – it happens automatically; check that *Slå på* wasn’t pressed
Crackling	Raise the DAW’s buffer size, or lower Unison

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